COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 9

RELATIONS PROPERTIES REPRESENTATION

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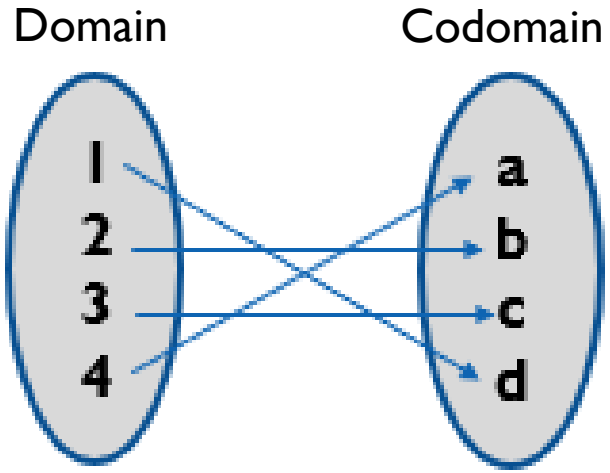






RELATIONS

- The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. For this reason, sets of ordered pairs
  are called *binary relations*.



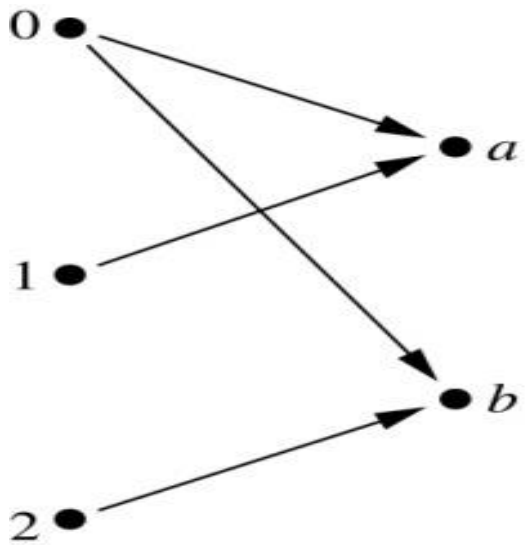
- We can use relations to solve problems involving communications networks, project scheduling, and identifying elements in sets with common properties.

- Binary Relations: Let A and B be sets. A *binary relation from A to B* is a subset of $A \times B$.
  In other words, a binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B.

- We use the notation $a\ R\ b$ to denote that $(a, b) \in R$. When (a, b) belongs to R, a is said to be related to b by R.
- *Note*: $a\ R\ b$ means a is not related to b by R, i.e., $(a, b) \notin R$.

RELATIONS

- Let $A = \{0,1,2\}$ and $B = \{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B. This means, for instance, 0 R a , but that 1 $\not R$ b
- Relations can be represented graphically or using a table.

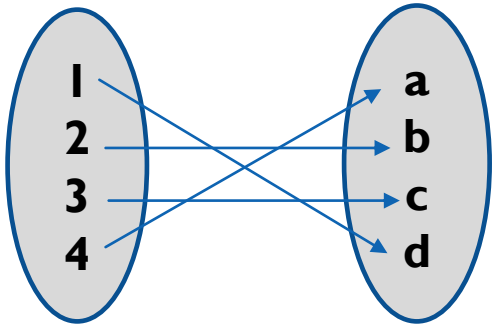


R	a	b
0	×	×
1	×	
2		×

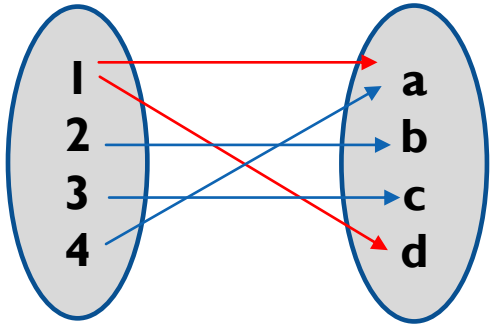
Note: If a relation is given as a table, the **domain** consists of the members of the first column and the **range** consists of the members of the second column.

FUNCTIONS VS RELATIONS

- A *relation* can be used to express a *one-to-many relationship* between the elements of the sets *A* (domain) and *B* (codomain), where an element of *A* (domain) may be related to more than one element of *B* (codomain).
- A *function* represents a relation where exactly one element of *B* (codomain) is related to each element of *A* (domain).
- Relations are more general than functions. A function is a relation where exactly one element of *B* is related to each element of *A*.



Function



Relations

RELATIONS ON A SET

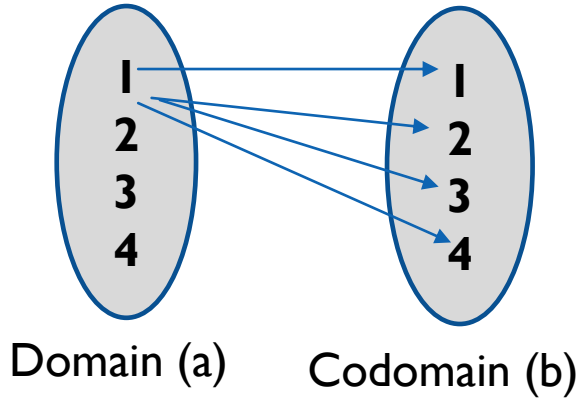
Definition: A *relation on a set A* is a relation from *A to A*. In other words, a relation on a set *A is a* subset of $A \times A$.

Example: Suppose that $A = \{a, b, c\}$. Then $R = \{(a, a), (a, b), (a, c)\}$ is a relation on A.

Q. Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution: Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b (a | b which is b / a), we see that

a | b (i.e. b/a) → 1/1 = 1, 2/1=2, 3/1=3, 4/1=4
1/2=0.5, 2/2=1, 3/2=1.5, 4/2=2
1/3=0.3, 2/3=0.6, 3/3=1, 4/3=1.3
1/4=0.25, 2/4=0.5, 3/4=0.75, 4/4=1



So, the relation would be the set where integer results are found

R = {(1,1), (1, 2), (1,3), (1, 4), (2, 2), (2, 4), (3, 3),(4, 4)}

Here in each set, the first element is from domain and second element is from codomain

RELATIONS ON A SET

Q: How many relations are there on a set with n elements?

Solution: A relation on a set A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$. Thus, there are 2^{n^2} relations on a set with n elements.

For example, there are $2^{3^2} = 2^9 = 512$ relation on the set $\{a, b, c\}$

PROPERTIES OF RELATIONS: *REFLEXIVE* RELATION

- Definition: A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$
- Using quantifiers, a relation on the set A is reflexive if $\forall a\ ((a, a) \in R)$, where universe of discourse is the set of ALL elements in A.
- In a reflexive relation, every element is related to itself. i.e. $a\ R\ a$ for all $a \in A$
- *Example:* The relation $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3)\}$ on the set $\{1, 2, 3\}$, is reflexive.

Q: Is the "***divides***" relation on the set of **positive integers** reflexive?

Solution: Yes. Because $a \mid a$ whenever a is a positive integer, the "divides" relation is reflexive.

Q: Is the "***divides***" relation on the set of **integers** reflexive?

Solution: No. Because $0 \mid 0$ (0 does not divide 0)

PROPERTIES OF RELATIONS: *REFLEXIVE* RELATION

Q: Consider the following relations on {1, 2, 3, 4}:

R_1 = {(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)}

R_2 = {(1,1), (1,2), (2,1)}

R_3 = {(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)}

R_4 = {(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)}

R_5 = {(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)},

R_6 = {(3,4)}.

Which of these relations are reflexive?

Solution: The relations R_3 and R_5 are reflexive because they both contain ALL pairs of the form (a, a), namely (1,1) (2,2), (3,3) and (4,4).

PROPERTIES OF RELATIONS: *REFLEXIVE* RELATION

- Example: The following relations on the set of integers are ***reflexive***:

$R_1 = \{(a, b) \mid a \leq b\}$,

$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$,

$R_4 = \{(a, b) \mid a = b\}$.

The following relations are NOT reflexive:

$R_2 = \{(a, b) \mid a > b\}$ (note that $3 \not> 3$),

$R_5 = \{(a, b) \mid a = b + 1\}$ (note that $3 \neq 3 + 1$),

$R_6 = \{(a, b) \mid a + b \leq 3\}$ (note that $4 + 4 \not\leq 3$).

SYMMETRIC AND *ANTISYMMETRIC* RELATION

- Symmetric: A relation R on a set A is called *symmetric* if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.
- If (a, b) exists in a set then there will also (b, a) in the set. For example if there is (1, 3) then there will be also (3, 1) in the set.
- The relation $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,3), (1, 4)\}$ on *the set* $\{1, 2, 3, 4\}$ is symmetric.
- Antisymmetric: A relation R on a set A such that for all $a, b \in A$ if $(a, b) \in R$ and $(b, a) \in R$, then $a = b$ is called *antisymmetric*.
- If (a, b) exists in a set then there will be no (b, a) in the set except if there is only a = b. For example if there is (1, 3) then there will be no (3, 1) in the set.
- In other words, R is antisymmetric if whenever a = b, then $a \not R\, b$ or $b \not R\, a$.
- It follows that R is not antisymmetric if we have a and b in A, $a \neq b$, and both $a\ R\ b$ or $b\ R\ a$.
- Note: The terms *symmetric* and *antisymmetric* are NOT opposite, because a relation can have both of these properties or may lack both of them. For example $R = \{(1,1), (2,2), (3,3), (4,4)\}$

SYMMETRIC AND *ANTISYMMETRIC* RELATION

Example: The following relations on the integers are

Symmetric:

$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\}$

$R_4 = \{(a, b) \mid a = b\}$

$R_6 = \{(a, b) \mid a + b \leq 3\}$

The following are NOT symmetric:

$R_1 = \{(a, b) \mid a \leq b\}$ (note that $3 \leq 4$, but $4 \not\leq 3$)

$R_2 = \{(a, b) \mid a > b\}$ (note that $4 > 3$, but $3 \not> 4$)

$R_5 = \{(a, b) \mid a = b + 1\}$ (note that $4 = 3 + 1$, but $3 \neq 4 + 1$)

SYMMETRIC AND *ANTISYMMETRIC* RELATION

Q: Consider the following relations on {1, 2, 3, 4}:

$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$

$R_2 = \{(1,1), (1,2), (2,1)\}$

$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$

$R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$,

$R_6 = \{(3,4)\}$.

Which of the relations are *symmetric and which are antisymmetric*?

Solution: The relations R_2 *and* R_3 *are* symmetric.
The relations R_4 , R_5 , and R_6 are antisymmetric.

Q: What about R_1? Solution: neither symmetric not antisymmetric

TRANSITIVE RELATION

- Definition: A relation R on a set A is called *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.
- If there is (a, b) and (b, c) then there is also (a, c) in the set. For example, If there is (1, 3) and (3, 4) then there is also (1, 4) in the set if it is transitive
- *Example* : The relation $R = \{(1,1), (1,2),(1, 3), (1, 4), (2,2), (2,3), (2, 4), (3, 3), (3,4), (4, 4)\}$ on *the set* $\{1, 2, 3, 4\}$ is *transitive*.

Q. Is the relation $R = \{ (a, a), (b, c), (c, b), (d, d)\}$ on the set $X = \{a, b, c, d\}$ is ***transitive***?

Solution: No. Because (b, c) and (c, b) are in R, but (b, b) is not in R

TRANSITIVE RELATION

Q: Consider the following relations on {1, 2, 3, 4}, Which of the relations are ***transitive***?

$R_1 = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,1), (4,4)\}$ $R_2 = \{(1,1), (1,2), (2,1)\}$

$R_3 = \{(1,1), (1,2), (1,4), (2,1), (2,2), (3,3), (4,1), (4,4)\}$ $R_4 = \{(2,1), (3,1), (3,2), (4,1), (4,2), (4,3)\}$

$R_5 = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$, $R_6 = \{(3,4)\}$.

- ***Solution***: **R_4 , R_5 & R_6 : transitive** $\Leftarrow$ verify that if (a, b) and (b, c) belong to this relation then (a, c) belongs also to the relation. R_4 transitive since (3,2) and (2,1), (4,2) and (2,1), (4,3) and (3,1), and (4,3) and (3,2) are the only such sets of pairs, and (3,1) , (4,1) and (4,2) belong to R_4. Same reasoning for R_5 and R_6.
- **R_1 : not transitive** $\Leftarrow$ (3,4) and (4,1) belong to R_1, but (3,1) does not.
- **R_2 : not transitive** $\Leftarrow$ (2,1) and (1,2) belong to R_2, but (2,2) does not.
- **R_3 : not transitive** $\Leftarrow$ (4,1) and (1,2) belong to R_3, but (4,2) does not.

COMBINING RELATION

- Because relations from ***A*** to ***B*** are subsets of ***A X B***, two relations from ***A*** to ***B*** can be combined in any way two sets can be combined.
- Given two relations R_1 and R_2, we can combine them using basic set operations to form new relations such as $R_1 \cup R_2$, $R_1 \cap R_2$, $R_1 - R_2$, and $R_2 - R_1$.
- **Example**: Let $A = \{1,2,3\}$ and $B = \{1,2,3,4\}$. The relations $R_1 = \{(1,1),(2,2),(3,3)\}$ and $R_2 = \{(1,1),(1,2),(1,3),(1,4)\}$ can be combined using basic set operations to form new relations:

- $R_1 \cup R_2 = \{(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)\}$ (relations both in R1 and R2)
- $R_1 \cap R_2 = \{(1,1)\}$ (common relations in R1 and R2)
- $R_1 - R_2 = \{(2,2),(3,3)\}$ (relations exist in R1 but not in R2)
- $R_2 - R_1 = \{(1,2),(1,3),(1,4)\}$ (relations exist in R2 but not in R1)

COMPOSITE OF RELATIONS

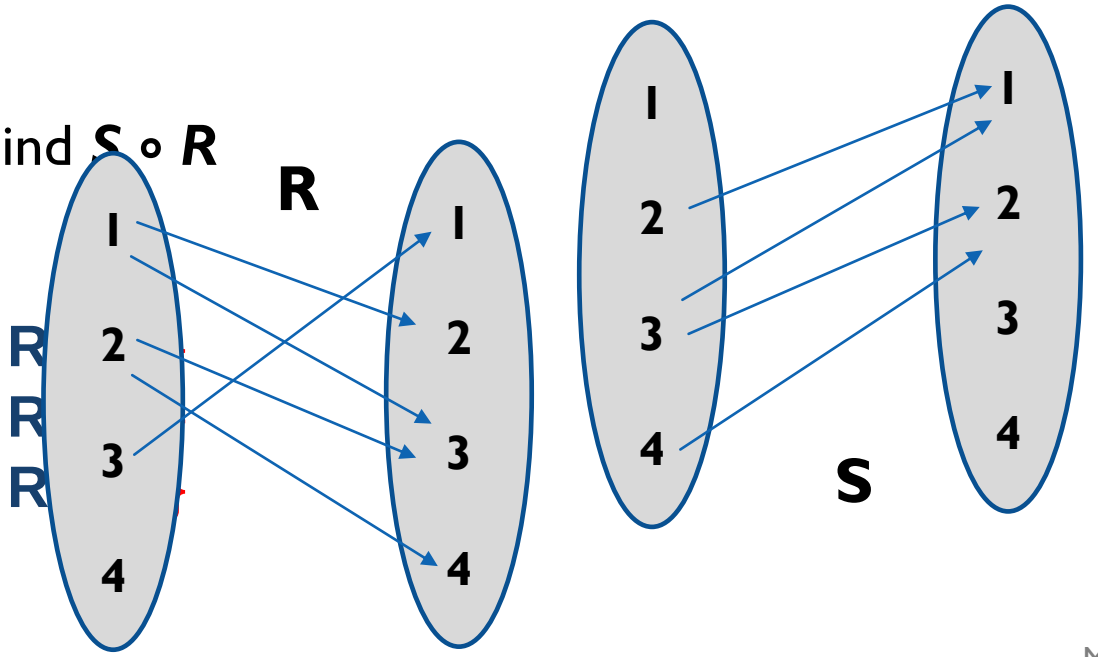
- Let ***R*** be a relation from *A* to *B* and ***S*** a relation from *B* to a set *C*. The composite of *R* and *S* is the relation consisting of ordered pairs *(a, c)*, where $a \in A$, $c \in C$, and for which there exists an element b $\in B$ such that $(a,b) \in R$ and $(b, c) \in S$. **Composite of *R* and *S*** denoted by ***S* ∘ *R***

- ***Note:*** Computing the composite of two relations requires that we find elements that are the **second elements** of ordered pairs in the **first relation**, and **first element** of ordered pairs in the **second relation**

Q. Let *R* be the relation {(1,2), (1,3),(2,3), (2,4),(3,1)}, and let *S* be the relation {(2,1), (3,1),(3,2), (4,2)}. Find ***S* ∘ *R***

Solution: ***S* ∘ *R* =** {(1,1), (1,2), (2,1), (2,2)}

R{1,2) & S{2,1} **S ∘ R** {1,1}; R{1,3) & S{3,1} **S ∘ R**
R{1,3) & S{3,2} **S ∘ R** {1,2}; R{2,3) & S{3,1} **S ∘ R**
R{2,3) & S{3,2} **S ∘ R** {2,2}; R{2,4) & S{4,2} **S ∘ R**
R{3,1) & S{ } **S ∘ R**

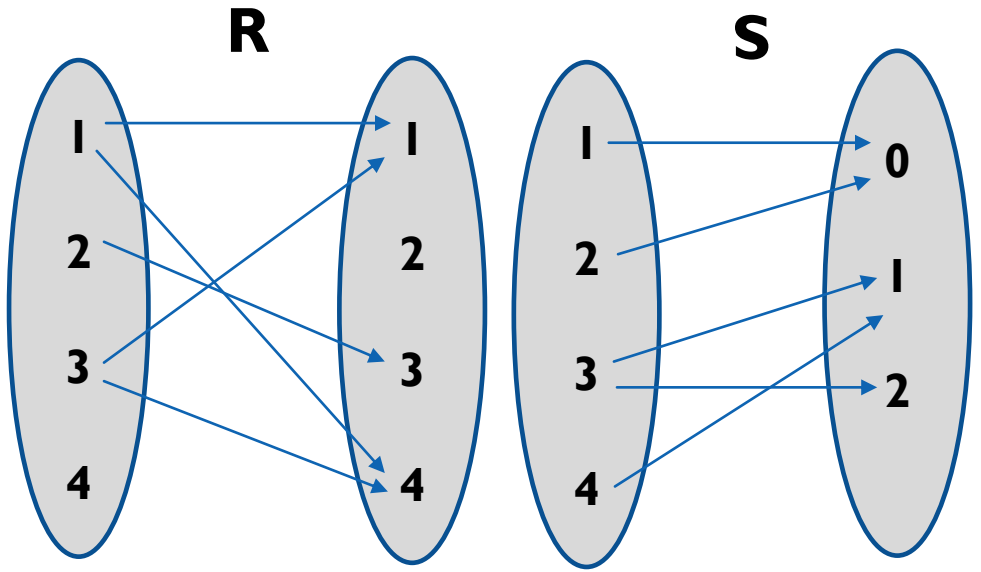


COMPOSITE OF RELATIONS

Q: What is the composite of the relations ***R*** and ***S,*** where ***R*** is the relation from {1, 2, 3} to {1, 2, 3, 4} with ***R*** = {*(1,1), (1,4), (2,3), (3,1), (3,4)*} and ***S*** is the relation from {*1, 2, 3, 4*} to {*0, 1, 2*} with ***S*** = {*(1,0), (2,0), (3,1), (3,2), (4,1)*}?

Solution: $S \circ R = \{ (1,0), (1,1), (2,1), (2,2), (3,0), (3,1) \}$

R{1,1) & S{1,0} **S ∘ R** {1,0}
R{1,4) & S{4,1} **S ∘ R** {1,1}
R{2,3) & S{3,1} **S ∘ R** {2,1}
R{2,3) & S{3,2} **S ∘ R** {2,2}
R{3,1) & S{1,0} **S ∘ R** {3,0}
R{3,4) & S{4,1} **S ∘ R** {3,1}



REFERENCES

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- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley